

Process Monitoring Dashboard

1. Problem Statement:

Develop a GUI-based tool to monitor system processes and resource usage, providing real-time insights into CPU, memory, disk, and network performance.

2. Key Features:

- Real-Time Monitoring: Track active processes, CPU, memory, disk, and network usage.
 - Process Management: View, sort, and terminate processes.
 - Resource Visualization: Graphical representation of system resource usage.
 - System Information: Display system stats such as kernel version and uptime.
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3. Tools & Platforms:

- Programming Language: Python
 - GUI Framework: Tkinter or PyQt
 - Libraries:
 - psutil (for process and resource data)
 - matplotlib (for graphs)
 - Platform: Windows, Linux
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4. Implementation Approach:

- Setup Environment:
 - Install Python.
 - Install Tkinter or PyQt for GUI development.
 - Install required libraries:

```
pip install psutil matplotlib
```
- Design UI:
 - Create a dashboard with sections for:
 - Process list
 - Resource graphs (CPU, memory, etc.)
 - System information
- Fetch System Data:
 - Use psutil to gather real-time data:
 - CPU and memory usage
 - Disk I/O and network activity
 - Active process details

- **Data Visualization:**

- Use matplotlib to plot graphs.
- Display data in tables for easy interpretation.

- **Process Management:**

- Implement options to:
 - Terminate processes.
 - Sort processes based on resource usage.

- **Testing:**

- Test the dashboard on different system loads and conditions.
- Ensure the application runs smoothly on Windows and Linux.

5. Typical Workflow:

- **Launch Application:**

- Start the dashboard from a Python script.

- **Real-Time Monitoring:**

- Automatically update process and resource data at regular intervals.

- **Interaction:**

- Allow users to:
 - Sort processes.
 - Terminate resource-heavy processes.

- **Resource Visualization:**

- Update graphs and tables periodically.

- **Exit:**

- Safely close processes and clean up resources.

6. Conclusion:

The Process Monitoring Dashboard aims to be a practical and user-friendly tool for system administrators and users, helping them to efficiently monitor and manage their system processes and resources. By providing real-time data and management options, this tool enhances system performance monitoring and decision-making.